



Intelligent Medicine and Prescription Reminder System

Software Design and Architecture
Project



Navigating the Medication Maze

1

The Challenge

Medication non-adherence is a silent crisis, affecting millions globally.

2

The Impact

Severe health risks, increased hospitalizations, and reduced quality of life.

3

The Vulnerable

Elderly and chronic patients face heightened risks due to complex regimens and memory decline.



The Problem: Medication Non-Adherence Crisis

Cardiac Patients

41% of heart attack patients miss antihypertensive medications.

Diabetic Patients

33% of diabetics regularly forget their doses.

Consequences

Missed or incorrect intake leads to severe health risks and increased hospitalizations.

Bridging the Adherence Gap



A Critical Need

The demand for reliable medication management solutions is urgent. Our system addresses this directly.

Who Benefits?

Patients, caregivers, and healthcare providers seeking to improve adherence and health outcomes.



Our Solution: Intelligent Medicine & Prescription Reminder System

01

Smart & User-Friendly

Intuitive system reminds patients of medication schedules accurately.

02

Integrated Tracking

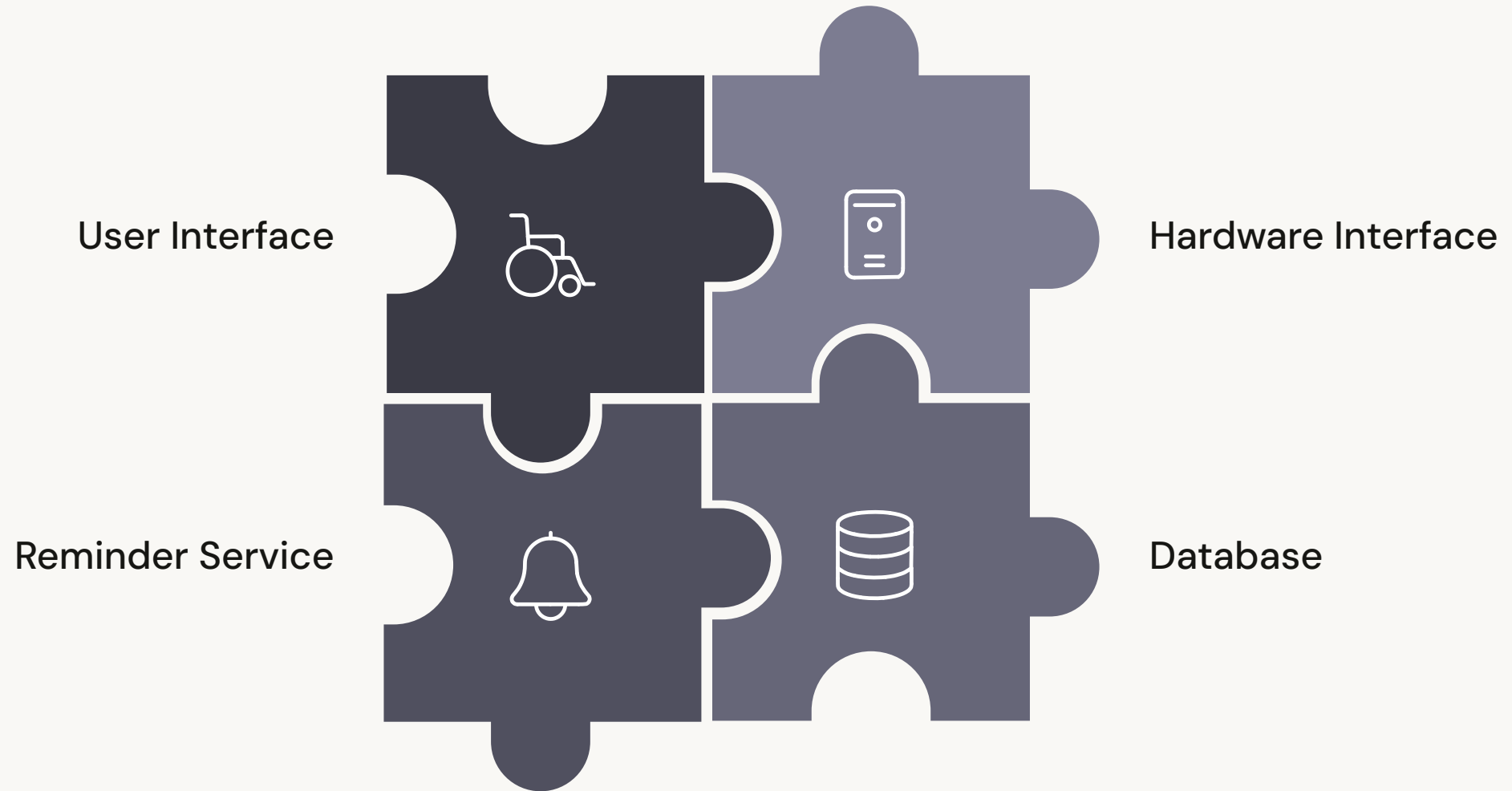
Combines software notifications with hardware for real-time adherence tracking.

03

Personalized Alerts

Supports dosage, timing, and prescription management with customizable alerts and escalation.

System Architecture Overview



This high-level component diagram illustrates the core modules and their interactions within our system.

Functional Requirements (FR) Overview

1

Schedule Management

Input, update, and securely store comprehensive medication regimens.

2

Reminder Alerts

Timely notifications via app, SMS, or calls, ensuring no dose is missed.

3

Adherence Tracking

Automatic logging of medication intake and missed doses for insights.

4

User Roles

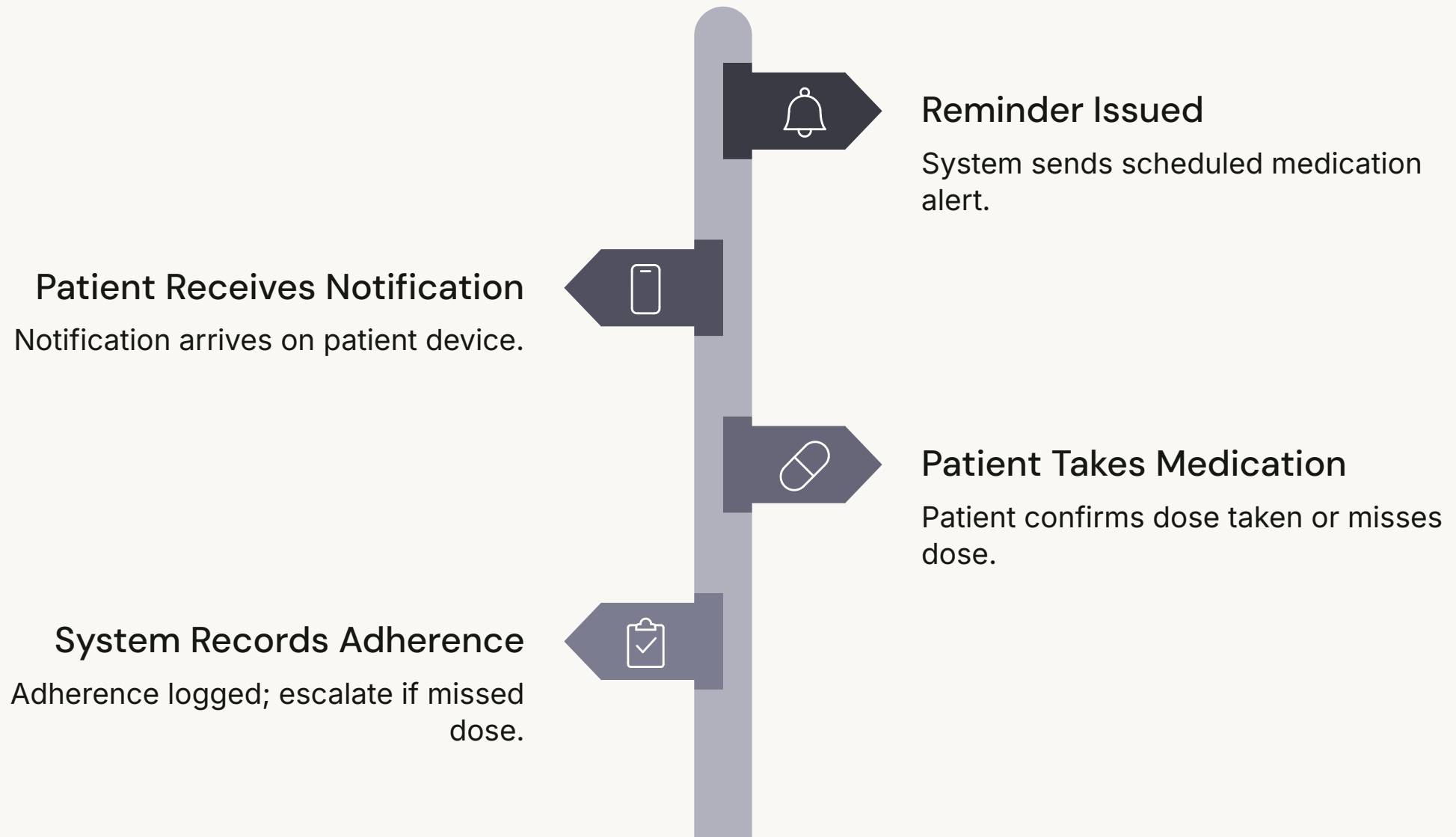
Dedicated interfaces for patients, caregivers, and healthcare providers.

5

Data Synchronization

Secure cloud storage and seamless multi-device access for all users.

Activity Diagram: Medication Taking Workflow



This activity diagram outlines the patient's interaction with the system from reminder to adherence logging.

Non-Functional Requirements (NFR) Highlights



Usability

Intuitive UI for elderly and non-tech-savvy users, simple and clear.



Reliability

99.9% uptime for reminders and alerts, guaranteeing critical function.



Security

HIPAA-compliant data encryption and stringent access control measures.



Scalability

Supports thousands of users with minimal latency, ready for expansion.

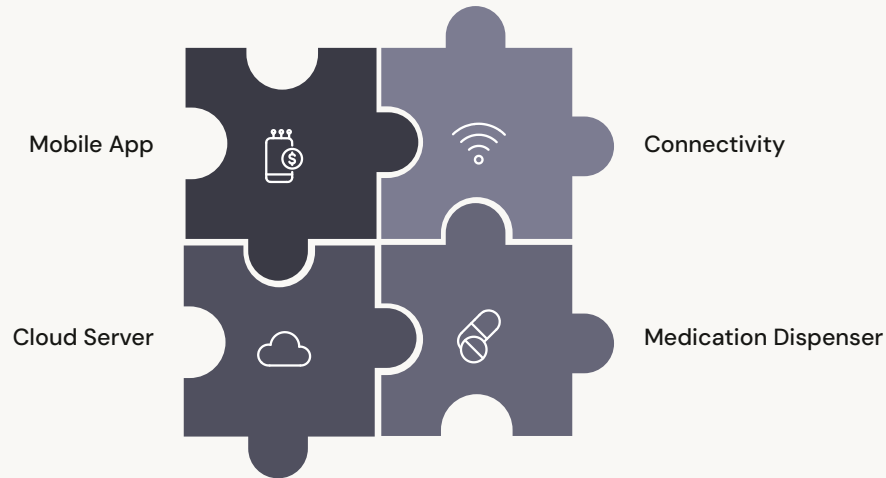


Maintainability

Modular design for easy updates, rapid feature additions, and long-term support.

Deployment & Class Diagram Snapshots

System Deployment



Visualizing how our system components are distributed across different environments.

Key Class Relationships



Illustrating the structure and relationships of critical data entities.